Huawei Access Node BTM-ACC-23 QoS Congestion Event Troubleshooting Procedure

Node: BTM-ACC-23 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_6970

Description

The BTM-ACC-23 access node located at the BTM Layout Aggregation site is equipped with Huawei access equipment, which utilizes a Quality of Service (QoS) framework to manage and prioritize traffic flows. In the event of a QoS congestion event, characterized by microbursts, interface tail drops, and QoS queue starvation, the network experiences reduced performance and increased latency. This is primarily due to the inefficient allocation of bandwidth resources, leading to packet loss and increased latency. The Huawei QoS framework, based on the Differentiated Services (DiffServ) model, relies on the classification, policing, and queuing of traffic flows to ensure optimal network performance. However, when the QoS configuration is misaligned with the network's traffic patterns, congestion events can occur, resulting in suboptimal network performance. The ALARM_6970 alarm code, indicating a minor severity QoS congestion event, necessitates prompt troubleshooting to identify the root cause and implement corrective measures.

Pre-requisite Checks

- Verify that the BTM Layout Aggregation site is accessible and that all necessary permissions have been obtained.

- Confirm that the redundant link status is normal and that there are no planned maintenance activities in progress.

- Ensure that the Huawei access equipment is powered on and that all fans are operational.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Insufficient bandwidth allocation	Medium
Interface congestion due to high traffic load	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy, Expected Output: Display QoS policy configuration, Description: This command displays the QoS policy configuration, including the classification, policing, and queuing settings. The output will help identify any misconfigured policies that may be contributing to the QoS congestion event.

Command: display interface, Expected Output: Display interface status, Description: This command displays the status of the interfaces, including the congestion level and packet loss statistics. The output
```

```
will help identify any interfaces that are experiencing congestion and may be contributing to the QoS congestion event.

Command: display qos queue statistics, Expected Output: Display QoS queue statistics, Description: This command displays the QoS queue statistics, including the queue depth, packet loss rate, and latency. The output will help identify any queues that are experiencing congestion and may be contributing to the QoS congestion event.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policy configuration, Command: config qos policy , Description: Verify the QoS policy configuration and adjust it as necessary to ensure that it is aligned with the network's traffic patterns. This may involve modifying the classification, policing, and queuing settings to ensure that traffic flows are properly prioritized and allocated bandwidth.

Step 2: Step: Increase bandwidth allocation, Command: config interface bandwidth , Description: Increase the bandwidth allocation for the affected interface to ensure that it has sufficient resources to handle the traffic load. This may involve modifying the interface configuration to allocate additional bandwidth or modifying the QoS policy to allocate more bandwidth to the affected traffic flows.

Step 3: Step: Optimize interface configuration, Command: config interface congestion-control , Description: Optimize the interface configuration to prevent congestion. This may involve modifying the congestion control settings or modifying the QoS policy to prioritize traffic flows and prevent congestion.

Post-Check Verification

```
Command: display qos policy, Expected Output: Verified QoS policy configuration, Description: Verify that the QoS policy configuration has been adjusted as necessary to ensure that it is aligned with the network's traffic patterns.

Command: display interface, Expected Output: Verified interface status, Description: Verify that the interface status has improved and that congestion has been resolved.

Command: display qos queue statistics, Expected Output: Verified QoS queue statistics, Description: Verify that the QoS queue statistics have improved and that congestion has been resolved.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: undo config qos policy , Description: Restore the original QoS policy configuration to ensure that the network returns to its previous state.

- Step: Restore original interface configuration, Command: undo config interface bandwidth , Description: Restore the original interface configuration to ensure that the network returns to its previous state.

- Step: Restore original congestion control settings, Command: undo config interface congestion-control , Description: Restore the original congestion control settings to ensure that the network returns to its previous state.